

ATHARVA JITENDRA KHAIRE

Data Scientist | Data Analyst | Data Engineer

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Portfolio: <https://deepsix-pi.vercel.app>

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GitHub: <https://github.com/beastlyluck>

PROFESSIONAL SUMMARY

Master of Data Science candidate at Monash University seeking a Data Scientist, Data Analyst, or Data Engineer role. Builds end-to-end analytics: data wrangling, SQL, statistical modelling, machine learning, time series forecasting, causal inference, ETL data pipelines, and interactive dashboards. Portfolio of six digital twins and 22 analytics case studies using Python, Pandas, NumPy, scikit-learn, PySpark, PyTorch, Tableau, FastAPI, Docker, and AWS. Focused on data quality, feature engineering, model evaluation, SHAP explainability, KPI reporting, and numbers stakeholders can audit.

TECHNICAL SKILLS

Programming: Python, SQL, R, Java, C++, JavaScript, TypeScript, Linux shell, Git, GitHub, CI/CD (GitHub Actions)

Data analysis: Pandas, NumPy, data wrangling, exploratory data analysis (EDA), statistical modelling, hypothesis testing, A/B testing, causal inference, time series forecasting, feature engineering, model evaluation, data quality

Machine learning: scikit-learn, XGBoost, LightGBM, supervised learning, classification, regression, clustering, anomaly detection, SHAP, model monitoring, drift detection, PyTorch, computer vision, NLP

Data engineering: ETL, data pipelines, Apache Spark / PySpark, dbt-style modelling, Great Expectations, Parquet, PostgreSQL, FastAPI, REST APIs, Docker, Kubernetes, AWS, Kafka / Avro

Visualization and BI: Tableau, Power BI, Plotly, Matplotlib, ggplot2, Streamlit, dashboard design, KPI reporting, GeoPandas

Working style: Agile delivery, stakeholder communication, project management, documentation, code review

SELECTED PROJECTS

Portfolio: <https://deepsix-pi.vercel.app> | Source: <https://github.com/beastlyluck>

AeroTwin - Digital twin (aerospace) | Python, PyTorch, TorchScript

- Trained a drone control policy in simulation, then scored sim-to-real gap on held-out wind. Six airframes, 400 Hz edge loop, domain randomisation vs a held-out world. GitHub: github.com/beastlyluck/aerotwin

GridPulse - Digital twin (energy) | Python, SciPy, HiGHS, SQL-style streaming

- Built a 48-bus microgrid twin: DC power flow, physics-informed line rating, and LP dispatch so one overloaded line does not cascade. Compared nameplate vs derated schedules. github.com/beastlyluck/gridpulse

BioSync - Digital twin (wearables) | Python, NumPy, ONNX, Streamlit

- Continuous-time glucose and heart-rate replica that keeps running when the wearable drops out. RK4 ODE, missingness masks, on-device ONNX runtime, differential-privacy cohort export. github.com/beastlyluck/biosync

OceanicOS - Digital twin (logistics) | Python, NetworkX, FastAPI

- Discrete-event container terminal: 2 berths, 16 AGVs, Hungarian assignment, 48-hour policy comparison so the yard does not jam. Hierarchical volume forecasts reconciled with MinT. github.com/beastlyluck/oceanicos

ForgeX - Digital twin (manufacturing) | Python, OpenCV, scikit-learn

- Late-fusion cell twin: acoustics, current, and vision. IsolationForest plus logistic score, HOLD/ESTOP interlock that will not self-reset. Scored on a later shift than the fit set. github.com/beastlyluck/forgeo

TerraTwin - Digital twin (agriculture) | Python, NumPy, Docker, Kubernetes

- Closed-loop climate for an 8-tower vertical farm. Fitted Q-iteration vs PID on the same seed; spatial disease model; facility.toml as the config contract. github.com/beastlyluck/terratwin

Campus load forecast and model watch desk | Python, LightGBM, time series, drift monitoring

- Hourly electricity demand for 12 buildings: LightGBM + LSTM residual, rolling-origin evaluation, 24-hour MAPE 6.8%. PSI / Evidently drift gate so a winter model cannot silently fail in a heatwave.

Nightly analytics fabric and score API | Python, dbt, Great Expectations, FastAPI, XGBoost, SHAP

- ETL pipeline: 27 dbt-style models, 42 data-quality tests, 18-minute nightly SLA; failed tests block publish. Credit-style score API: 12 features, AUC 0.86, p95 latency 11 ms, top-3 SHAP reason codes, canary rollback.

Causal inference and forecasting cases | Python, R, SQL, Tableau

- Retention uplift (Qini 0.19 at 20% targeting, 34% budget saved at equal retention). Fare synthetic control (ATT -6.4% boardings, placebo tests). SKU hierarchy MinT (WRMSSE 0.62, 12k SKUs, coherence error 0). Ward occupancy twin: MAE 4.2 beds, 83% interval coverage.

Computer vision, graphs, and operations boards | Python, PyTorch, YOLOv8, NetworkX, Tableau

- EuroSAT land-use change: 97.4% accuracy, unsure tiles reported. Inspection KPI board: mAP 0.69, PPM -18%. Fraud-ring case packs: PR-AUC 0.64, review time -31%. Victorian ED wait board, rent-stress atlas, invoice-leak queue.

EXPERIENCE

Data Science Project Contributor - Mylan Laboratories

May 2022 - Jul 2022

Nashik, India | *Manufacturing analytics*

- Built a production-line dashboard for manufacturing insights using Python, Pandas, and data visualization.
- Analysed system efficiency metrics and presented findings to a cross-functional team for data-driven decisions.

Assistant Manager - ERTH Electric Pvt. Ltd.

Jun 2023 - Mar 2024

Nashik, India | *Full-time | Web, content, and project delivery*

- Managed the content and video pipeline for web and social channels; contributed to a successful website launch.
- Project-managed mobile application delivery and implemented frontend (HTML, CSS, JavaScript) with backend compatibility checks.

Volunteer - Swapnapurti Foundation

2020 - 2024

Nashik, India | *Community outreach*

- Ran technology workshops for students in remote village schools and supported community clean-up and outreach events.

EDUCATION

Master of Data Science - Monash University

2024 - 2026

Clayton, Melbourne, Australia | *Coursework complete; FIT5122 and FIT5120 in progress*

- Units include Python, SQL / databases, data wrangling, statistical modelling, machine learning, PySpark / big data, Tableau and data visualisation, deep learning (PyTorch), cloud computing and security (AWS, Docker, Kubernetes), project management, and IT research methods.
- Industry experience studio project (FIT5120) and professional practice (FIT5122) in progress. Agile delivery with an external client.

Bachelor of Computer Science - Savitribai Phule Pune University

Graduated Jan 2024

Pune, India

- Capstone: historical figures in a virtual environment using Unreal Engine 5, C++, and AI behaviour systems.

CERTIFICATIONS

- Learning Python for Data Analysis and Visualization - Udemy (Jun 2022) | Jupyter, NumPy, Pandas, Matplotlib
- Complete Blender Creator: 3D Modelling for Beginners - Udemy (Feb 2024)
- Unreal Engine 5 C++ Developer - Udemy (Feb 2024)

ADDITIONAL INFORMATION

Languages: English, Hindi, Marathi. Open to Data Scientist, Data Analyst, and Data Engineer roles in Melbourne.
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